Assignment1

Info

Name: Cody (Zhexian) Liu
Assignment Number: 1
Repo Link: https://github.com/Rorical/CWM-FDI

Basic Timing

What is the difference you observe between real, user, and sys time?

```
ubuntu@ubuntu:~/CWM-FDI/assignment1$ /usr/bin/time -v time python3 matmul_slow.py
n=128 reps=2 checksum=107389.290000
0.21user 0.00system 0:00.23elapsed 99%CPU (0avgtext+0avgdata 13308maxresident)k
164inputs+0outputs (0major+1945minor)pagefaults 0swaps
        Command being timed: "time python3 matmul_slow.py"
        User time (seconds): 0.21
        System time (seconds): 0.01
        Percent of CPU this job got: 99%
        Elapsed (wall clock) time (h:mm:ss or m:ss): 0:00.23
        Average shared text size (kbytes): 0
        Average unshared data size (kbytes): 0
        Average stack size (kbytes): 0
        Average total size (kbytes): 0
        Maximum resident set size (kbytes): 13308
        Average resident set size (kbytes): 0
        Major (requiring I/O) page faults: 0
        Minor (reclaiming a frame) page faults: 2030
        Voluntary context switches: 46
        Involuntary context switches: 5
        Swaps: 0
        File system inputs: 244
        File system outputs: 0
        Socket messages sent: 0
        Socket messages received: 0
        Signals delivered: 0
        Page size (bytes): 4096
        Exit status: 0
ubuntu@ubuntu:~/CWM-FDI/assignment1$
```

```
ubuntu@ubuntu:~/CWM-FDI/assignment1$ time python3 matmul_slow.py
n=128 reps=2 checksum=107389.290000

real    0m0.234s
user    0m0.224s
sys     0m0.009s
```

The system time is smallest among all the time measurements. The real time is the longest

because it measures the total amount of time the program spend. It is the sum of user and sys time.

Which summary statistics would you report: mean, median, or minimum? Why?

```
ubuntu@ubuntu:~/CWM-FDI/assignment1$ for i in 1 2 3 4 5; do /usr/bin/time -f "%e" python3 m
atmul_slow.py; done
n=128 reps=2 checksum=107389.290000
0.22
n=128 reps=2 checksum=107389.290000
0.22
n=128 reps=2 checksum=107389.290000
0.22
n=128 reps=2 checksum=107389.290000
0.22
n=128 reps=2 checksum=107389.290000
0.22
```

Mean. The mean time of execution represents the expectation of time the program costs, which will be the real and stable amount of time it takes when deployed in large scale.

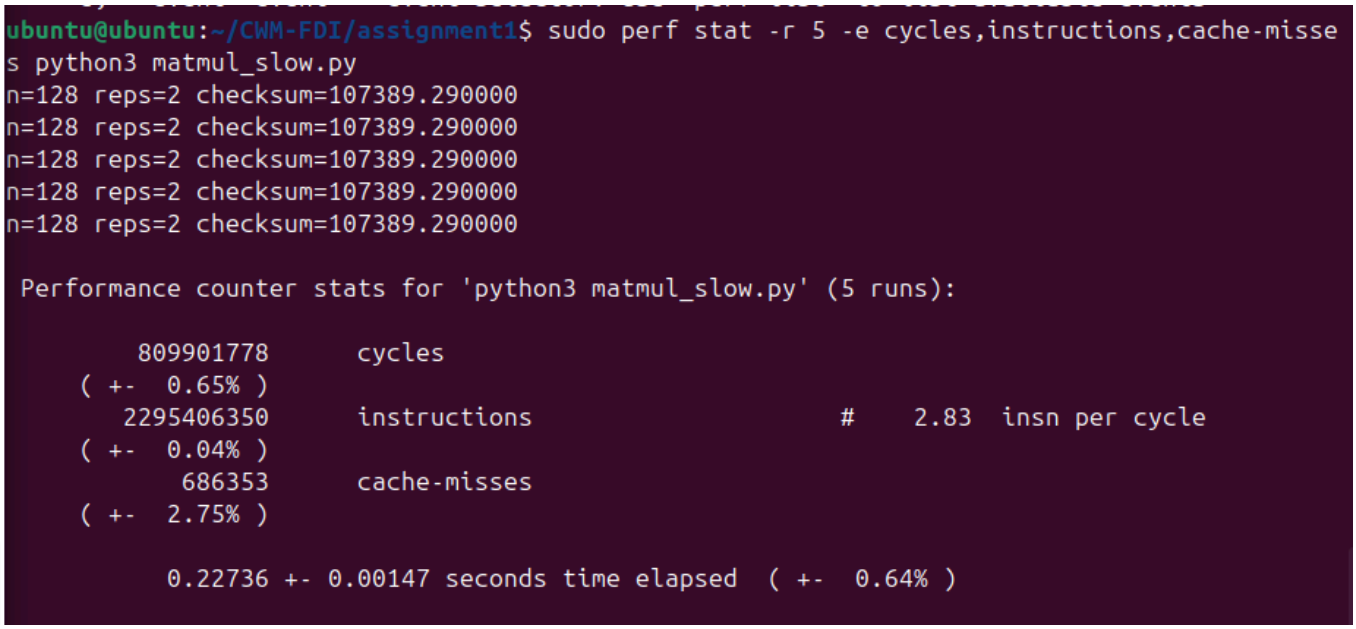
Reading Counters with perf stat

Report the output.

```
ubuntu@ubuntu:~/CWM-FDI/assignment1$ sudo perf stat -e cycles,instructions python3 matmul_s
low.py
n=128 reps=2 checksum=107389.290000

 Performance counter stats for 'python3 matmul_slow.py':

         817666898      cycles

        2299661070      instructions                     #    2.81  insn per cycle


       0.234871740 seconds time elapsed

       0.224805000 seconds user
       0.009032000 seconds sys
```

It reported 817666898 cycles; 2299661070 instructions with 2.81 instructions-per-cycle ratio

Report the instructions-per-cycle ratio. What does it suggest?
Report cache misses you observe. Why may this matter?



The perf shows 2.83 instructions per cycle. It suggests that CPU is efficiently executing multiple instructions in a single cycle. With that value higher, the program is more efficient and execute faster because it has higher CPU utilization.

Cache miss is 686353 times. This means CPU cannot find the data it want in caches, and has to fetch from memory or other places. CPU has three layers of caches that accelerate its memory reads. More cache miss means slower execution due to bandwidth limits between CPU and memory and each memory operation costs more time.

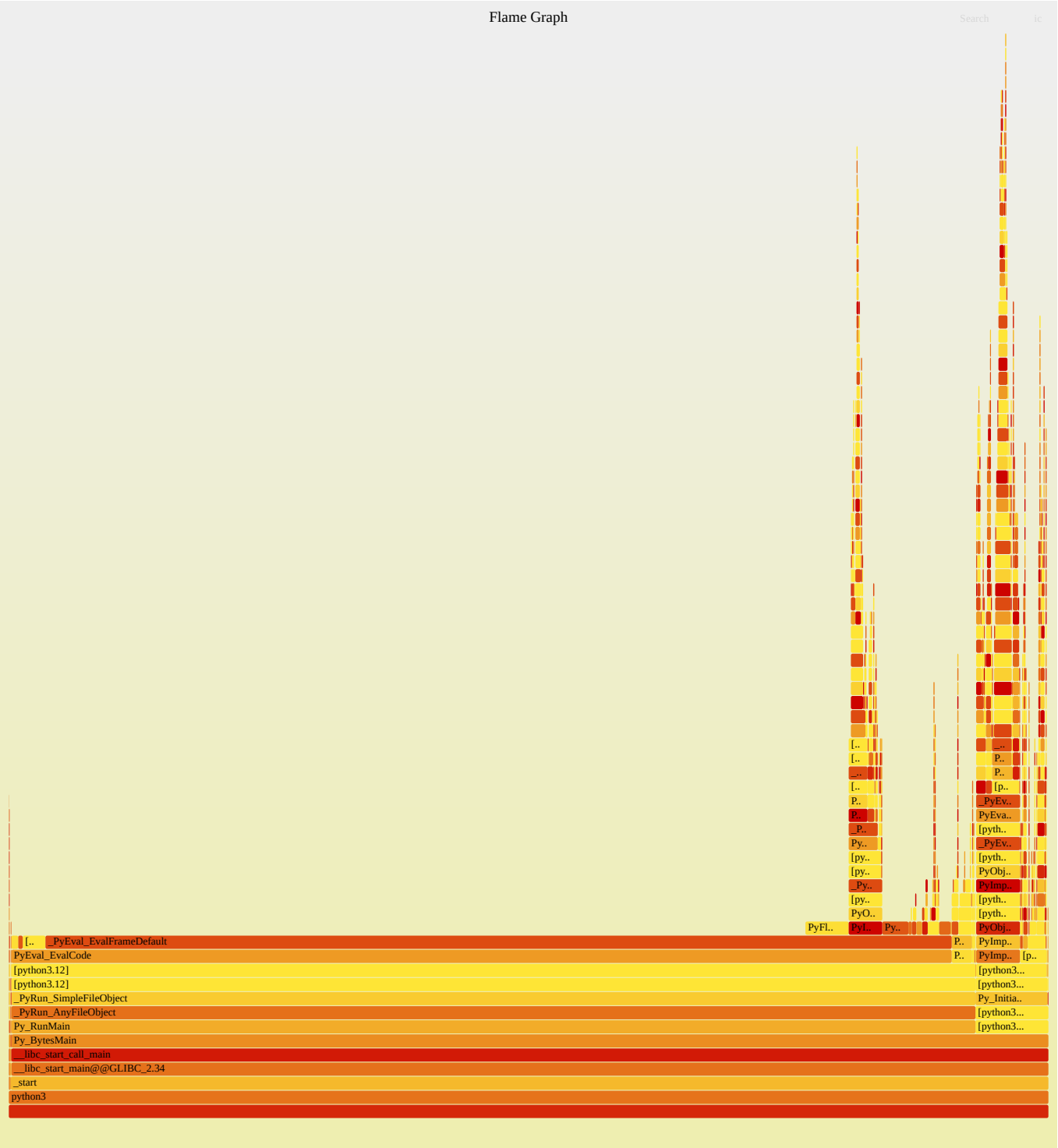
Hotspot discovery with perf record and perf report

```
Samples: 919  of event 'cycles:P', Event count (approx.): 811549601
Overhead  Command  Shared Object         Symbol
  75.47%  python3  python3.12           [.] _PyEval_EvalFrameDefault
   3.41%  python3  python3.12           [.] PyFloat_FromDouble
   3.19%  python3  python3.12           [.] PyLong_FromLong
   1.54%  python3  python3.12           [.] _Py_NewReference
   0.99%  python3  python3.12           [.] 0x0000000000156483
   0.55%  python3  python3.12           [.] 0x000000000015646f
   0.55%  python3  python3.12           [.] PyObject_Malloc
   0.44%  python3  python3.12           [.] PyObject_Free
   0.44%  python3  [kernel.kallsyms]    [k] __irqentry_text_end
   0.31%  python3  [kernel.kallsyms]    [k] strncmp
   0.30%  python3  [kernel.kallsyms]    [k] inflate_fast
   0.22%  python3  python3.12           [.] 0x0000000000156458
   0.22%  python3  python3.12           [.] 0x000000000015647a
   0.22%  python3  python3.12           [.] PyLong_AsDouble
   0.22%  python3  python3.12           [.] 0x0000000000021d940
   0.22%  python3  python3.12           [.] 0x0000000000021db64
   0.22%  python3  python3.12           [.] 0x0000000000021dbc4
   0.22%  python3  libc.so.6            [.] __memset_avx2_unaligned_erms
   0.22%  python3  python3.12           [.] _PyCode_New
   0.22%  python3  python3.12           [.] PyObject_IS_GC
   0.20%  python3  [kernel.kallsyms]    [k] clear_page_erms
   0.20%  python3  libc.so.6            [.] __memmove_avx_unaligned_erms
   0.18%  python3  [kernel.kallsyms]    [k] native_irq_return_iret
   0.16%  python3  [kernel.kallsyms]    [k] do_user_addr_fault
   0.13%  python3  [kernel.kallsyms]    [k] copy_mc_enhanced_fast_string
   0.11%  python3  python3.12           [.] PyLong_AsLongAndOverflow
   0.11%  python3  python3.12           [.] 0x0000000000015bbdf
   0.11%  python3  python3.12           [.] PyLong_AsLong
   0.11%  python3  [kernel.kallsyms]    [k] error_entry
   0.11%  python3  python3.12           [.] 0x0000000000016f6d2
   0.11%  python3  python3.12           [.] 0x0000000000018b74a
   0.11%  python3  python3.12           [.] 0x0000000000021dbf0
   0.11%  python3  python3.12           [.] 0x0000000000156e71
   0.11%  python3  python3.12           [.] 0x0000000000016a705
   0.11%  python3  python3.12           [.] 0x000000000015647d
   0.11%  python3  python3.12           [.] 0x0000000000021d320
   0.11%  python3  python3.12           [.] 0x0000000000166e15
   0.11%  python3  python3.12           [.] 0x0000000000021ced8
   0.11%  python3  python3.12           [.] 0x0000000000021db89
   0.11%  python3  python3.12           [.] 0x0000000000156daa
   0.11%  python3  python3.12           [.] 0x0000000000159c76
   0.11%  python3  [kernel.kallsyms]    [k] alloc_pages_mpol
```

Which function consumes the largest fraction of samples?

PyEval_EvalFrameDefault consumes 75% of overhead, but it is used for execution of python bytecode, which is not interesting. The second large consumption comes from **PyFloat_FromDouble** which means python creating floating point number objects. I guess that in doing the $O(n^3)$ operations to each element of matrix, it create new floating point for each one which slows down the program.

Flame Graphs



a. Which function dominates runtime?

PyEval_EvalFrameDefault

b. Given this dominating call, what are the likely bottlenecks in the matmul_slow() program?

PyEval_EvalFrameDefault is used to execute python codes. This domination means it spent most of time executing pure python code. Python is interpreter based and very slow to execute line by line. The bottleneck is that all the computationally intensive functions are in python. It is

better to use numpy or torch which dispatch the actual computation to C code and optimized hardware.

Propose optimizations for speeding up the execution of matmul_slow.py

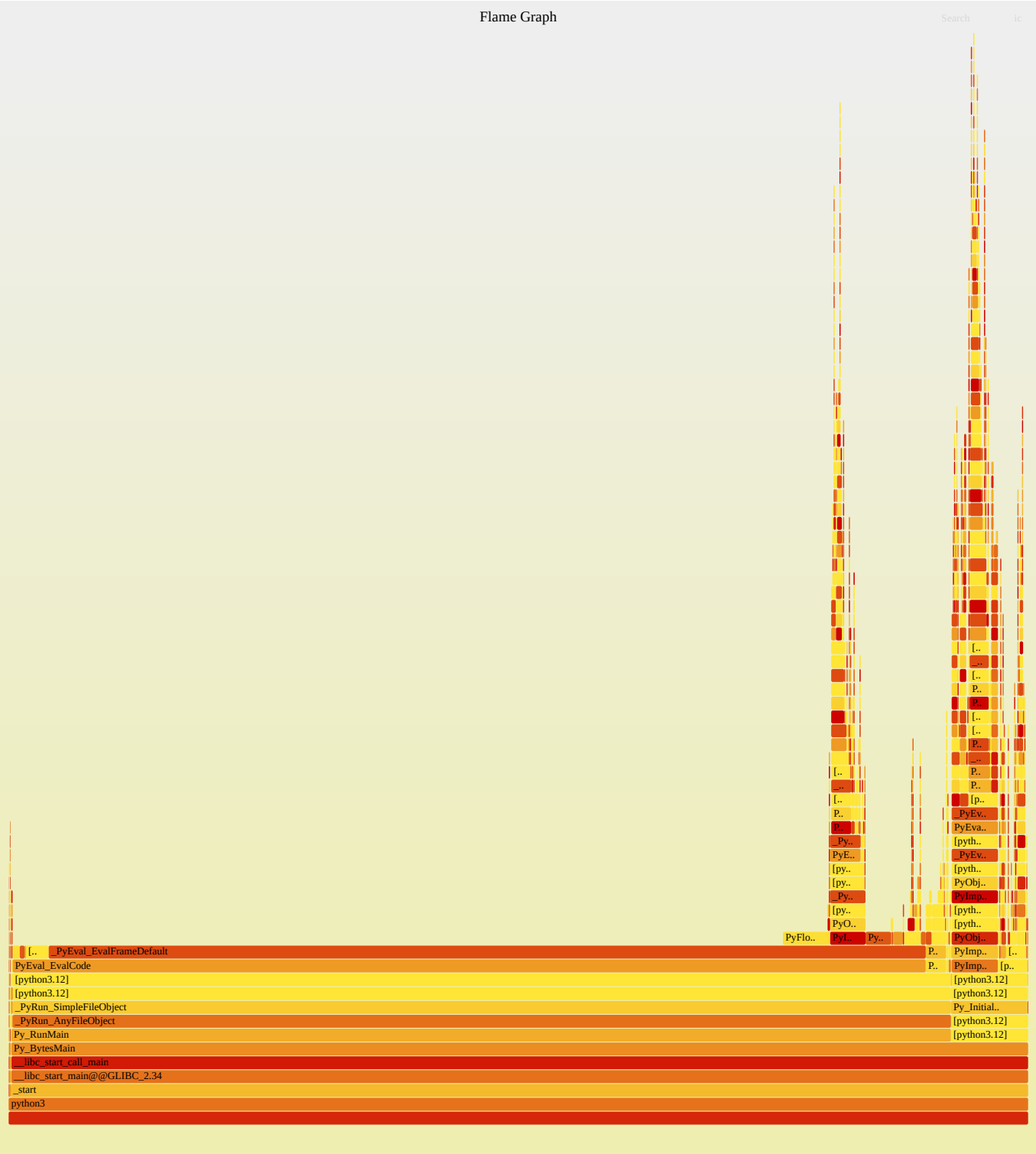
The matrix is defined as double nested list in the code:

```
Matrix = List[List[float]]
```

The slow version has no optimization on memory access pattern which hurts the cache.

Fast 1

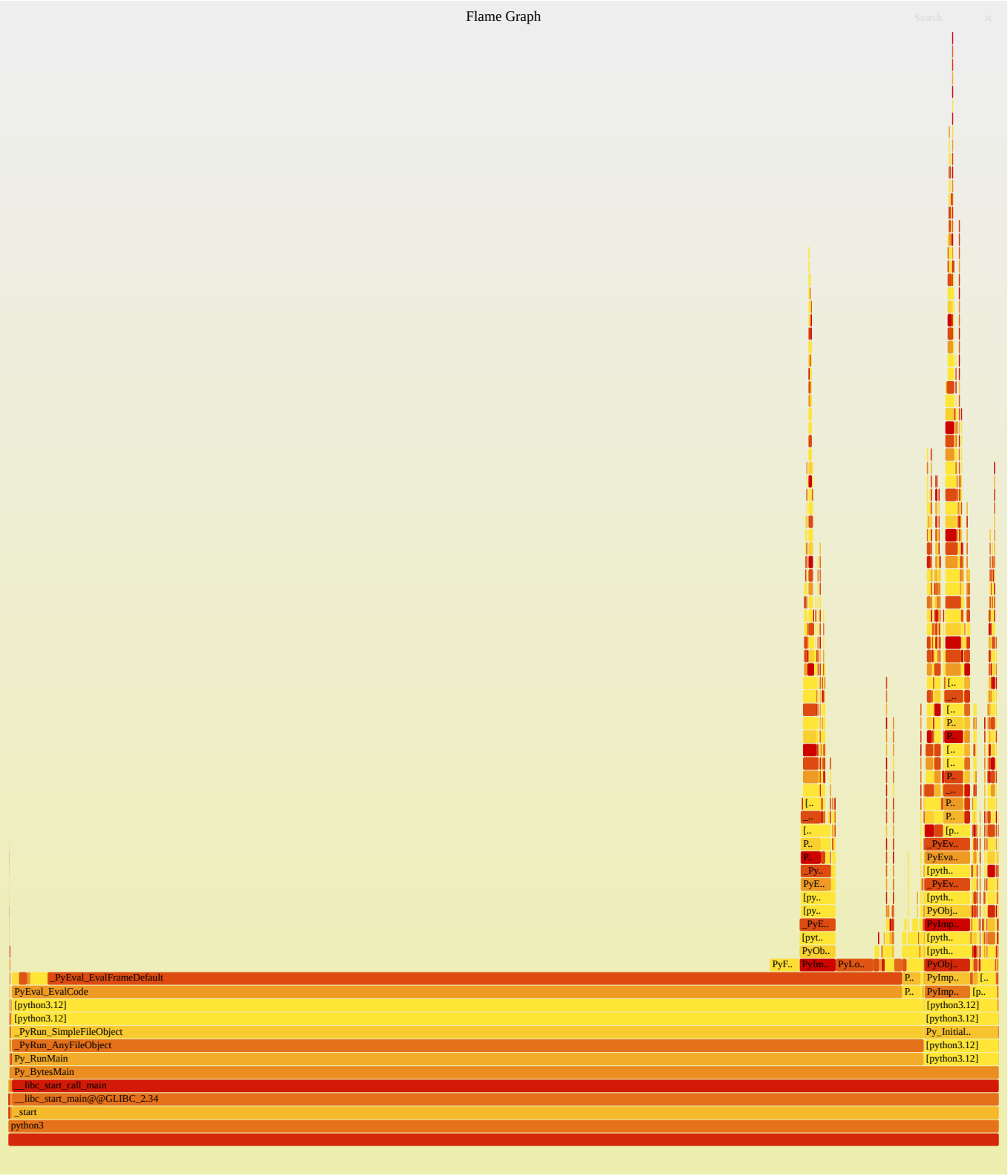
```
ubuntu@ubuntu:~/CWM-FDI/assignment1$ for i in 1 2 3 4 5; do /usr/bin/time -f "%e
" python3 matmul_fast.py; done
n=128 reps=2 checksum=107389.290000
0.21
n=128 reps=2 checksum=107389.290000
0.20
n=128 reps=2 checksum=107389.290000
0.20
n=128 reps=2 checksum=107389.290000
0.20
n=128 reps=2 checksum=107389.290000
0.20
ubuntu@ubuntu:~/CWM-FDI/assignment1$ sudo perf stat -r 5 -e cycles,instructions
python3 matmul_fast.py
n=128 reps=2 checksum=107389.290000
n=128 reps=2 checksum=107389.290000
n=128 reps=2 checksum=107389.290000
n=128 reps=2 checksum=107389.290000
n=128 reps=2 checksum=107389.290000

 Performance counter stats for 'python3 matmul_fast.py' (5 runs):

         733747640      cycles
                ( +-  0.21% )
        2060380304      instructions              #    2.81  insn per cyc
le              ( +-  0.05% )

           0.20872 +- 0.00330 seconds time elapsed  ( +-  1.58% )
```



Fast 1 version improve some speed by making locality access of arrays. It first make references of a and c matrices of their rows and then operate of the rows reference directly without accessing rows independently in that loop.

Fast 2

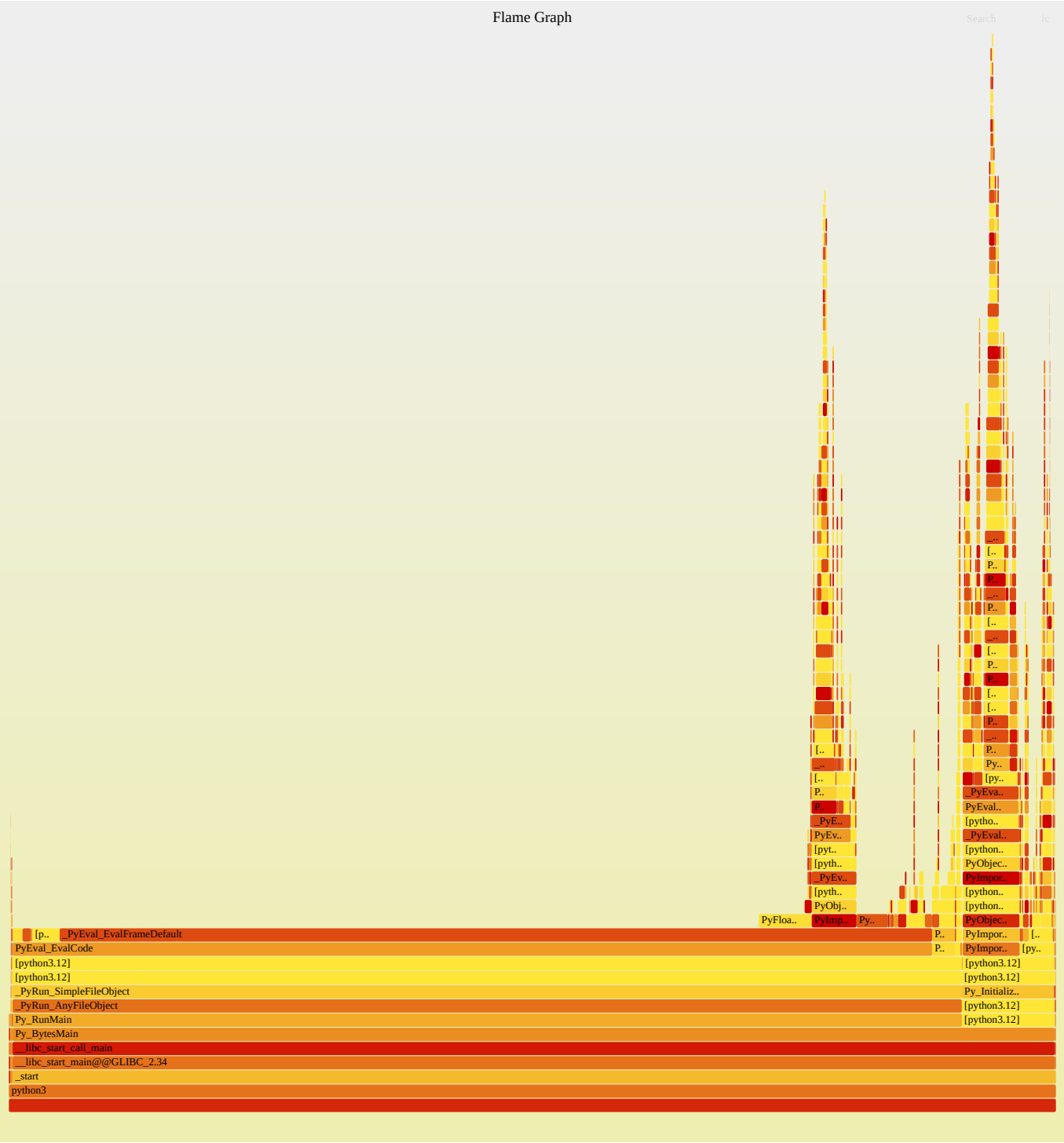
```
ubuntu@ubuntu:~/CWM-FDI/assignment1$ for i in 1 2 3 4 5; do /usr/bin/time -f "%e
" python3 matmul_fast.py; done
n=128 reps=2 checksum=107389.290000
0.21
n=128 reps=2 checksum=107389.290000
0.21
n=128 reps=2 checksum=107389.290000
0.21
n=128 reps=2 checksum=107389.290000
0.21
n=128 reps=2 checksum=107389.290000
0.21
ubuntu@ubuntu:~/CWM-FDI/assignment1$ sudo perf stat -r 5 -e cycles,instructions
python3 matmul_fast.py
n=128 reps=2 checksum=107389.290000
n=128 reps=2 checksum=107389.290000
n=128 reps=2 checksum=107389.290000
n=128 reps=2 checksum=107389.290000
n=128 reps=2 checksum=107389.290000

 Performance counter stats for 'python3 matmul_fast.py' (5 runs):

         759084385      cycles
                ( +-  0.59% )
        2273354841      instructions              #    2.99  insn per cyc
le              ( +-  0.04% )

          0.21389 +- 0.00256 seconds time elapsed  ( +-  1.20% )
```



Fast version 2 removed the repeated assignment of variable `total` inside the loop, by overriding matrix c to 0 and accumulate directly on it. This saves one memory variable creation and access.

Fast 3

```
ubuntu@ubuntu:~/CWM-FDI/assignment1$ for i in 1 2 3 4 5; do /usr/bin/time -f "%e
" python3 matmul_fast.py; done
n=128 reps=2 checksum=107389.290000
0.21
n=128 reps=2 checksum=107389.290000
0.17
n=128 reps=2 checksum=107389.290000
0.17
n=128 reps=2 checksum=107389.290000
0.18
n=128 reps=2 checksum=107389.290000
0.18
ubuntu@ubuntu:~/CWM-FDI/assignment1$ sudo perf stat -r 5 -e cycles,instructions
python3 matmul_fast.py
n=128 reps=2 checksum=107389.290000
n=128 reps=2 checksum=107389.290000
n=128 reps=2 checksum=107389.290000
n=128 reps=2 checksum=107389.290000
n=128 reps=2 checksum=107389.290000

 Performance counter stats for 'python3 matmul_fast.py' (5 runs):

         651556763      cycles
                ( +-  0.64% )
        1837531308      instructions                     #    2.82  insn per cyc
le              ( +-  0.04% )

           0.18284 +- 0.00107 seconds time elapsed  ( +-  0.58% )
```



Fast version 3 is most useful in improving efficiency by changing the memory accessing pattern completely. Instead of doing column access on matrix b for every point, it first take transpose then access every rows of b. This makes memory access predictable and adjacent, so cache works the best. The computation is same, but CPU runs faster on more cache access.